8.2 Detector Types

A Tour of Every Major Detector Technology

Mann, Chapter 8 · Rewritten with extended explanations and figures

The Design Problem

Section 8.1 told us *how* particles deposit energy. This section asks: **how do we catch that energy and turn it into a measurable signal?** Every detector technology exploits one or more of the energy-loss mechanisms from 8.1, but each makes different trade-offs between spatial resolution, time resolution, energy resolution, triggering speed, and cost.

Key Vocabulary Refresher

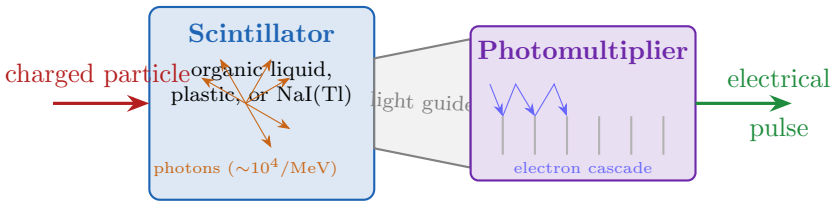
Before we dive in, here are terms that recur throughout:
Ionization: knocking electrons free from atoms. This is the primary signal in most detectors.
Scintillation: a material that emits *visible light* when a charged particle excites its atoms. The light is the signal.
Photomultiplier tube (PMT): a device that converts a single photon into a measurable electrical pulse via cascading electron emission at a series of electrodes (dynodes). Amplification: $\sim 10^4$–10^9.
Triggering: deciding in real time whether an event is "interesting" enough to record. A fast detector that can trigger saves you from recording terabytes of boring data.
Spatial resolution: how precisely you can locate the particle's trajectory (in microns or mm).
Time resolution: how precisely you can determine *when* the particle arrived (in nanoseconds or picoseconds).

8.2.1 Scintillation Counters

What they are

The oldest detector technology, dating to 1903 (Crookes' spinthariscope: a ZnS screen + the human eye). Modern versions replace the eye with a photomultiplier tube.

How they work



1. Charged particle enters the scintillator material and excites atoms/molecules.
2. Atoms de-excite by emitting **photons** (UV → converted to blue by dye molecules in plastics).
3. Photons travel through a light guide to a **photomultiplier tube**.
4. At the PMT photocathode, photons liberate electrons (photoelectric effect, ∼25% efficiency).
5. Electrons cascade through 6–14 **dynodes** at increasing voltage, amplifying by $\sim 10^4$–10^9.
6. The output pulse goes through a **discriminator** (rejects thermal noise) and a **pulse-height analyzer** (digitizes the signal).

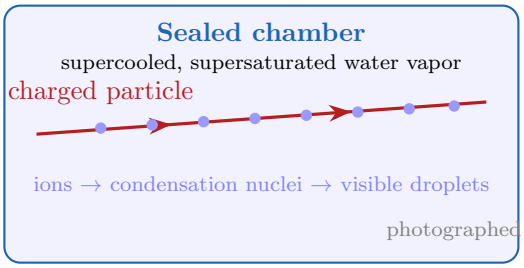
Two flavors of scintillator

NaI(Tl) (sodium iodide doped with thallium): high detection efficiency, but slow pulse decay (∼250 ns). Good for precise energy measurement.
Plastic scintillators (anthracene, naphthalene): fast pulse decay (∼10 ns, resolution ∼200 ps). Ideal as **trigger detectors**—their speed decides whether to activate the rest of the apparatus. Lower efficiency.

8.2.2 Cloud Chambers

How they work

Invented by C.T.R. Wilson (c. 1911) for studying rain cloud formation—then repurposed for particle physics.



The medium is at the edge of condensation. Ionization by a charged particle creates ions that serve as **nucleation centers**—tiny seeds around which vapor condenses into visible droplets. The result: a trail of mist along the particle's path.

Historical importance

Anderson used a cloud chamber in 1933 to discover the **positron**—the first antiparticle ever observed. The track curvature in a magnetic field showed a particle with the electron's mass but opposite charge.

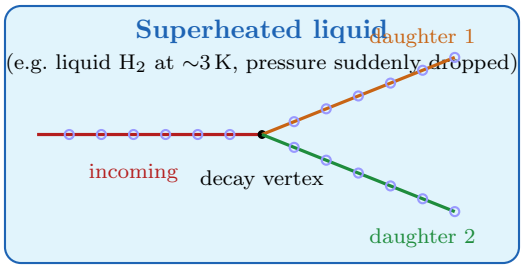
Nucleation center

A site (ion, dust speck, impurity) where a phase transition can begin. In a supersaturated vapor, ions left by a passing particle are ideal nucleation centers—droplets form around them preferentially.

8.2.3 Bubble Chambers

Invented by Glaser (1952). The bubble chamber is the *inverse* of a cloud chamber: instead of droplets forming in vapor, **bubbles form in superheated liquid**.

Operating principle



1. Start with very cold liquid under ~5 atm pressure.
2. Just before the beam arrives, rapidly **drop the pressure** (expand volume by ~1%). The liquid is now *superheated*—hotter than its boiling point at this pressure.
3. A charged particle ionizes the liquid. Ions are excellent nucleation centers for **boiling**, so bubbles form along the track.
4. Photograph the bubble trails. In a magnetic field, track curvature gives momentum via $pc = qBr$.

Strengths and weaknesses

Strength: sees *everything*—all charged particles, all decay vertices, full topology of complex multi-particle events. The $K^- + p \to \Omega^- + K^+ + K^0$ discovery photograph is a famous example. **Weaknesses:** (1) Cannot be triggered—it photographs every event like a surveillance camera, so every frame must be scanned. (2) Cannot be used at colliders because the beam geometry is incompatible with the chamber design. (3) Slow cycle time.

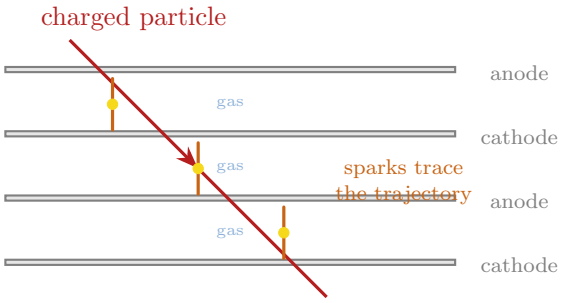
8.2.4 Spark Chambers

Electrical breakdown

When a sufficiently strong electric field is applied across a gas, free electrons accelerate enough to ionize gas molecules, which liberate more electrons, causing an **avalanche**—a cascade of ionization that creates a conducting "spark" channel.

How they work

Parallel plate electrodes with a gas between them. A short, intense high-voltage pulse (10–50 kV/cm) is applied *after* a charged particle passes through. The pre-existing ion trail from the particle seeds localized breakdowns—**sparks** that trace out the trajectory.



Advantage over bubble chambers

Spark chambers are **triggerable**: you can use auxiliary detectors (e.g. scintillation counters) to decide whether an event is interesting *before* firing the HV pulse. This makes them ideal for searching for rare events—you only photograph what matters. Data can be collected electronically rather than photographically.

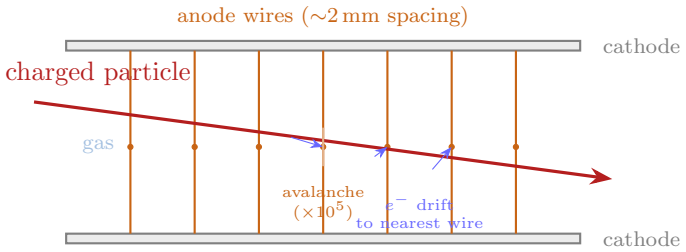
8.2.5 Wire Chambers (MWPCs)

Developed by Georges Charpak (1968, Nobel Prize 1992). The workhorse of modern tracking.

Key terms

Anode: positive electrode (the wires). **Cathode:** negative electrode (aluminum foil planes). **Avalanche:** exponential multiplication of electrons near a high-field region. **Drift chamber:** a wire chamber where you measure the *time* it takes electrons to drift to the wire, giving position from the drift distance.

How they work



1. A charged particle ionizes the gas, producing electron-ion pairs.
2. Freed electrons **drift** toward the nearest anode wire under the applied field.
3. Near the wire, the field is intense ($\propto 1/r$). Electrons gain enough energy to ionize more gas $\rightarrow$ **avalanche** with amplification $\sim 10^5$.
4. The wire that fires tells you *which* wire the particle was closest to $\rightarrow$ position.
5. **Drift chamber upgrade:** by timing how long the electron pulse takes to arrive (relative to an external trigger), you get the *drift distance*, yielding position resolution up to **100 μm**.
6. **Momentum measurement:** place MWPC planes on either side of a magnetic field region. The track deflection gives p via $pc = qBr$.

Sandwich geometry

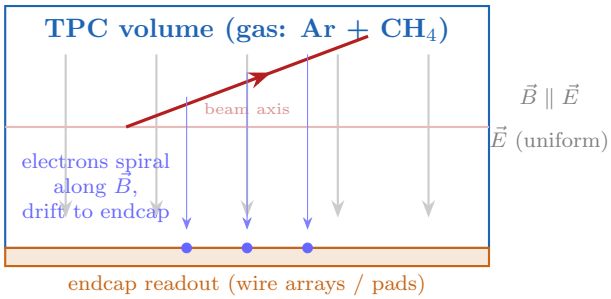
Stacking multiple wire planes at different angles lets you reconstruct full 3D trajectories. This is the basis of all modern tracking systems.

8.2.6 Time Projection Chambers (TPCs)

Invented by David Nygren (1974). The TPC gives **full 3D tracking** over a large solid angle—it is the ultimate tracking detector.

What makes it different from a wire chamber

In a wire chamber, you get 2D from wire positions and have to stack planes for 3D. A TPC gives all three coordinates from a *single* gas volume: two from where the electrons hit the endcap, and the third from *when* they arrive.



1. A charged particle ionizes the gas, leaving a trail of electron-ion pairs.
2. The uniform $\vec{E}$ field drifts the electrons toward one endcap. The parallel $\vec{B}$ field forces them into tight helical paths along the field axis—this prevents lateral diffusion and preserves spatial information.
3. **Two coordinates** come from *where* the electrons hit the endcap readout pads.
4. The **third coordinate** comes from *when* they arrive (drift time $\times$ drift velocity = distance from endcap).
5. The **total charge** deposited gives $dE/dx \rightarrow$ particle identification.

STAR detector at RHIC

The TPC in the STAR experiment at Brookhaven tracks *thousands* of particles simultaneously from a single heavy-ion collision—something no other detector type can do as effectively.

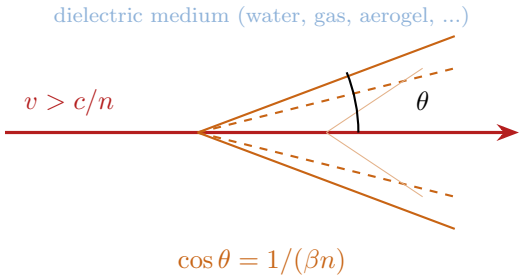
8.2.7 Cherenkov Counters

Cherenkov radiation refresher

When a charged particle moves through a dielectric medium *faster than light in that medium* ($v > c/n$, where n is the refractive index), it emits a coherent cone of light—analogous to a sonic boom. The emission angle satisfies:

$$\cos\theta = \frac{1}{\beta n}$$

Threshold: no radiation if $\beta < 1/n$. This threshold is what makes Cherenkov counters useful for particle ID.



Particle identification trick

Different particles with the *same momentum* p have different velocities β (because they have different masses). A Cherenkov counter with a given refractive index n will produce a signal only if $\beta > 1/n$.

Example: 1 GeV/c particles

From $p = m\gamma v$, the velocities at $p = 1\,\text{GeV}/c$ are:

Particle	Mass (GeV/c^2)	β	Needs $n >$
π	0.14	0.99	1.01
K	0.49	0.89	1.12
p	0.94	0.73	1.37

Two-detector trick: use water ($n = 1.33$) and a gas ($n = 1.03$) in sequence. The pion fires both, the kaon fires only the water detector, and the proton fires neither. Three-way separation from two detectors!

The **RICH** (Ring-Imaging Cherenkov) detector improves this by measuring the *angle* θ directly: the Cherenkov photons hit a detector plane and form a **ring** whose radius encodes θ and hence β.

Cherenkov photon yield (Mann Eq. 8.26)

$$\frac{dN}{d\ell\, d\lambda} = 2\pi\alpha\left(1 - \frac{1}{\beta^2 n^2}\right)\frac{1}{\lambda^2}$$

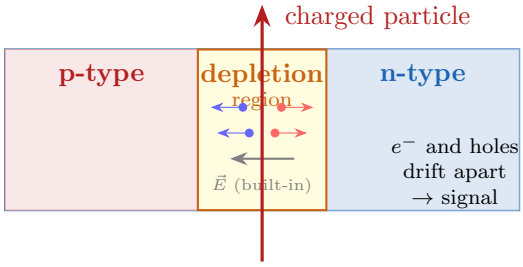
Typical yield: ∼200–300 photons/cm. Feeble! Detectors must be meters long.

8.2.8 Solid-State (Silicon) Detectors

Semiconductor refresher

In a semiconductor (e.g. silicon), the energy gap between the valence and conduction bands is small (∼1 eV). A passing charged particle can easily promote electrons across this gap, creating **electron-hole pairs**. In silicon, creating one pair costs only ∼3 eV—much less than the ∼30 eV needed to ionize a gas molecule. This means much better energy resolution.
A **p-n junction** (diode) has a **depletion region**: a zone with no free carriers, maintained by the built-in electric field from ionized dopant atoms. This is where detection happens.

How they work



1. A charged particle traverses the depletion region (∼300 μm thick).
2. It creates electron-hole pairs along its path (∼3 eV per pair).
3. The built-in electric field separates the pairs: electrons drift one way, holes the other.
4. This produces a current pulse whose position corresponds to the particle's trajectory.

Silicon microstrips

By fabricating the diodes as **parallel strips** spaced ∼25 μm apart, you get ∼5–10 μm resolution. Gluing two strip planes at an angle gives **two coordinates** per crossing → full 3D tracking.

Why silicon dominates modern tracking

Silicon microstrip (and pixel) detectors sit at the *innermost* layer of every modern collider detector (ATLAS, CMS, etc.). Their extreme spatial resolution allows identification of **secondary vertices**—displaced decay points of short-lived particles like B mesons and τ leptons. This capability was essential for discovering the top quark and the Higgs boson.

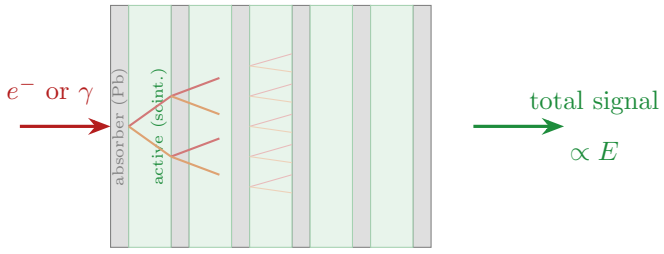
8.2.9 Calorimeters

Calorimeter

A detector that **absorbs all the kinetic energy** of an incident particle, converting it into a measurable signal (light, charge, etc.). Unlike tracking detectors, which try to disturb the particle as little as possible, a calorimeter *destroys* the particle to measure its energy.

How they work

The incident particle initiates a **shower** (cascade) of secondary particles. The total signal is proportional to the original energy.



There are two main types:

Sampling calorimeters: alternating layers of dense absorber (Pb, Fe) and active detector material (scintillator, liquid argon). Cheaper, but you only sample a fraction of the shower.

Homogeneous calorimeters: a single material that is both absorber and detector (e.g. lead glass, $PbWO_4$ crystals in CMS). Better resolution but much more expensive.

Energy resolution

$$\frac{\sigma_E}{E} \sim \frac{1}{\sqrt{E}}$$

The resolution *improves* at higher energies (because there are more particles in the shower, so statistical fluctuations average out). This is the opposite of tracking detectors, whose resolution degrades at high momentum.

Summary: Detector Comparison

Detector	Signal	Best at	Resolution	Triggerable
Scintillator	light from excitation	timing, triggering	∼200 ps	Yes
Cloud chamber	droplets in vapor	visual tracks	∼mm	No
Bubble chamber	bubbles in liquid	full event topology	∼mm	No
Spark chamber	sparks in gas	rare event search	∼mm	Yes
Wire chamber (MWPC)	avalanche on wire	tracking	∼100 μm	Yes
TPC	drift electrons	3D tracking, dE/dx	∼100 μm	Yes
Cherenkov	light cone	particle ID (β)	$\Delta\beta \sim 10^{-7}$	Yes
Silicon strip	e-hole pairs	precision vertex	∼5–10 μm	Yes
Calorimeter	shower energy	E measurement	$\sim 1/\sqrt{E}$	Yes

The historical arc

Notice the progression: cloud chambers and bubble chambers gave us beautiful photographs but couldn't be triggered or used at colliders. Wire chambers, TPCs, and silicon detectors are electronic, fast, triggerable, and layerable—exactly what you need at a modern collider where billions of events per second fly through the detector and only a handful are interesting.

How these fit into a modern collider detector (preview of 8.3)

A modern detector is an **onion**: layers arranged by what they catch.
Layer 1 (innermost): Silicon pixel/strip detectors → precision tracking, secondary vertices.
Layer 2: Drift/wire chambers in magnetic field → momentum from curvature.
Layer 3: EM calorimeter (high-Z, ∼25 X_0) → absorbs $e^{\pm}$ and γ completely.
Layer 4: Hadron calorimeter (∼10 interaction lengths) → absorbs hadrons.
Layer 5 (outermost): Muon chambers → only muons and neutrinos get this far. Neutrinos escape entirely; their presence is inferred from missing momentum.